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Predict Length of Stay and Optimize Bed Allocation

Frontline Forecast

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ABSTRACT

The ability to anticipate how long the patients are going to be admitted is critically affected by effective hospital bed management. This challenge is operationally consequential. This study presents a machine learning framework, developed in SAS Viya, to predict hospital length of stay among diabetic patients. In the dataset of 101,766 patients, we trained and evaluated various classification and regression models which helped us identify the strongest predictor as LightGBM for LOS prediction and we also used neural networks for readmission classification. The forecast was embedded in a 30-day bed allocation simulation with scenario-based planning for routine as well as urgent conditions. This framework provides a scalable, interpretable path towards data driven hospital operations management.

INTRODUCTION

Hospital bed-capacity planning hinges on accurate length-of-stay (LOS) forecasts; misestimation induces either overcrowding and care gaps or idle resources and escalating costs. Although machine-learning models trained on large patient corpora surpass traditional LOS predictors, most studies stop at prediction and omit operational translation.

We close this gap by (i) developing and benchmarking a suite of LOS models that output calibrated uncertainty intervals, (ii) embedding these predictions in a discrete-event bed-allocation simulation, and (iii) addressing three clinically actionable questions:

Which patient subgroups exhibit the highest LOS unpredictability?

Which admission-time features most strongly signal prolonged hospitalization?

Can encounter-level data alone reliably flag early discharge readiness?

DATA & FEATURES

Dataset Overview

The study employs the publicly available UCI Diabetes 130-US Hospitals dataset, comprising 101,766 hospital encounters of diabetic patients across 130 US hospitals. The dataset consists of demographic characteristics (age, race, gender), clinical indicators (primary and secondary diagnoses, medications, lab procedures), and healthcare utilization patterns (prior inpatient, outpatient, and emergency visits). The continuous length-of-stay variable (time_in_hospital, range: 1–14 days) serves as the major modeling target.

Exploratory Findings

The statistical analysis of the eight main numerical features showed a very strong right skew in data utilization. Most of the patients had zero prior or urgent visits; however, some groups of patients created a long tail in the data. The LOS was moderately right skewed (skewness=1.12) with a median of 4 days and a mean of 4.4 days indicating longer hospital stays.

LOS showed moderate positive correlations with number of medications ($r = 0.47$) and number of lab procedures ($r = 0.32$) which led us to the answer that the patients receiving more intensive treatment tend to stay longer which supported the reason of including the features in the predictive model.

Demographically, Caucasian patients formed the largest group where females were more represented and most of the diabetic patients were in the age group 50-90 with known type 2 diabetes trends.

DATA PARTITIONING

The data were partitioned at the patient level using *GroupShuffleSplit* with patient_nbr to mitigate the data leakage. This approach ensured that all the encounters associated with one patient were contained in a single data partition. The resulting split allocated 70% of patients to the training set (71,520 records), 15% to the validation set (15,027 records), and 15% to the test set (15,219 records).

Two predictive targets were derived from the time_in_hospital variable. The first was a continuous regression target representing the exact length of stay (LOS) in days. The second was a binary classification target (los_long_ge_3), distinguishing long stays (≥ 3 days) from short stays (< 3 days).

BASELINE MODELING

Linear Regression and Poisson Regression were used as baseline models for LOS regression, where both models were trained on eight core numerical features. Linear Regression was fit on log-transformed LOS targets to stabilize variance, whereas Poisson Regression modeled LOS as a raw count outcome. The Linear Regression model achieved a test root mean squared error (RMSE) of 2.61 days and a mean absolute error (MAE) of 1.88 days. Poisson Regression yielded a slightly lower RMSE of 2.56 days; however, the model was still learning despite the iteration limits.

SAS VIYA MODELING

Machine Learning Model Development and Comparison

Advanced modeling was conducted following SAS® Viya® best practices for scalable ML pipelines. We evaluated four LOS regression models and three models, using the full preprocessed feature set across the patient-level partitions described above.

Length of Stay Regression

Four regression models were compared for exact LOS prediction:

- Linear Regression: RMSE = 2.61 days, MAE = 1.88 days (baseline)
- Poisson Regression: RMSE = 2.56 days, MAE = 1.94 days
- Neural Network (MLP, 64→32): Val MAE = 1.64 days, RMSE = 2.21 days; Test MAE = 1.63 days, RMSE = 2.19 days
- LightGBM Regression: Val MAE = 1.61 days, RMSE = 2.18 days, $R^2 = 0.477$; Test MAE = 1.60 days, RMSE = 2.15 days, $R^2 = 0.473$

LightGBM outperformed all comparators, explaining approximately 47% of variance in LOS and achieving the lowest error across both validation and test sets. Table 1 summarizes performance across all regression models.

Table 1. Length of Stay Regression Model Performance

Model	Test MAE (days)	Test RMSE (days)	R^2
Linear Regression	1.88	2.61	—
Poisson Regression	1.94	2.56	—
Neural Network (MLP)	1.63	2.19	—
LightGBM	1.60	2.15	0.473

Table 1. Comparison of LOS regression models on the held-out test set.

Long-Stay Classification

A binary classification task was defined to distinguish short stays (LOS < 3 days) from long stays (LOS ≥ 3 days), providing an interpretable operational risk signal. Three classification models were implemented:

- Logistic Regression (baseline)
- Gradient Boosting Classifier
- Neural Network Classifier (MLP, 64→32)

The Neural Network classifier achieved the strongest overall performance, with a test accuracy of 78.3% and ROC-AUC of 0.838. As shown in Table 2, performance was asymmetric: the model identified long-stay patients with substantially higher precision and recall (F1 = 0.844) than short-stay patients (F1 = 0.644), reflecting clearer complexity signals for prolonged hospitalizations at admission time.

Table 2. Long-Stay Classification Performance (Neural Network)

Class	Precision	Recall	F1-Score	Support
Short Stay (<3 days)	0.656	0.633	0.644	4,722
Long Stay (≥3 days)	0.838	0.851	0.844	10,497
Overall Accuracy	—	—	0.783	15,219
ROC-AUC	—	—	0.838	—

Table 2. Classification report for the Neural Network model (test set).

Uncertainty Estimation via Quantile Regression

Point predictions of LOS, however accurate on average, provide insufficient guidance for operational planning. A patient predicted to stay three days might leave in one or remain for seven. To address this, LightGBM quantile regression models were trained at the 10th, 50th (median), and 90th percentiles of the LOS distribution. The P50 model achieved a test MAE of 1.57 days and R^2 of 0.453.

Before calibration, the P10–P90 prediction interval achieved 75.8% empirical coverage on the validation set — below the 80% nominal target. Post-hoc interval scaling (expanding intervals symmetrically around P50 by a factor of 1.056) raised calibrated coverage to 79.5% on validation and 79.8% on the test set. The mean interval width on the test set was 4.76 days (median: 4.70 days), providing practically useful planning bounds for administrators.

Table 4. Quantile Regression Interval Statistics (Test Set)

Metric	Validation	Test
Raw P10–P90 Coverage	75.8%	76.0%
Calibrated Coverage	79.5%	79.8%
Mean Interval Width (days)	4.80	4.76
Median Interval Width (days)	4.75	4.70
P90 Interval Width (days)	7.24	7.18

Table 4. Pre- and post-calibration interval statistics across validation and test sets.

Subgroup Coverage Analysis

Calibrated uncertainty estimates were evaluated for consistency across demographic subgroups to identify potential equity concerns.

Gender: Coverage was 79.5% for female patients and 80.2% for male patients — a difference of less than one percentage point, indicating no systematic gender-based disparity in uncertainty estimation.

Age: Coverage ranged from 77.6% (ages 90–100) to 82.5% (ages 10–20), with all other age groups clustering between 78% and 81%. The modest variation across the lifespan suggests stable uncertainty behavior without severe under-coverage in any age cohort.

Race: Coverage was 79.9% for Caucasian, 79.8% for Hispanic, 79.3% for African American, and 80.0% for Asian patients. The 'Other' category recorded the lowest coverage at 76.4%, likely reflecting smaller sample size rather than systematic bias.

Overall, coverage estimates were robustly near the 80% nominal target across all reported subgroups, indicating that the calibration procedure generalized equitably across the patient population.

Unpredictability Analysis: Who is Hardest to Predict?

To characterize patients for whom LOS prediction is most challenging, we defined 'unpredictable' encounters as the top 10% of absolute prediction errors on the test set. The corresponding threshold was 3.50 days, yielding 1,522 high-error encounters out of 15,219 total.

Table 5. Feature Differences: Predictable vs. Unpredictable Patients

Feature	Predictable Mean	Unpredictable Mean	Difference
Lab Procedures	42.29	48.69	+6.41
Medications	15.63	19.01	+3.38
Age (years)	65.73	68.04	+2.31
Number of Diagnoses	7.36	7.91	+0.54
Procedures	1.30	1.62	+0.32
Prior Inpatient Visits	0.61	0.83	+0.22

Table 5. Mean feature values for predictable vs. unpredictable patient groups (top 10% absolute error threshold = 3.50 days).

These findings demonstrate that multimorbidity, treatment intensity, and prior utilization burden are strongly associated with elevated LOS unpredictability. Patients undergoing extensive diagnostic evaluation and complex pharmacologic management show greater discharge timing variability, likely reflecting evolving or unstable clinical trajectories. Notably, demographic variables showed minimal differences between groups, confirming that unpredictability is driven primarily by clinical complexity rather than patient demographics.

Proxy Discharge Readiness (LOS ≤ 2 Days)

True 24-hour discharge readiness prediction requires time-series clinical data — evolving laboratory values, vital sign trends, therapy clearance, nursing notes, and care progression flags — which are not available in the encounter-level dataset used here. As a proxy analysis, early discharge readiness was operationalized as LOS ≤ 2 days, and a binary MLP classifier was trained on admission-time features. On the held-out test set, the model achieved a test accuracy of 78.1% and ROC-AUC of 0.836. While these results demonstrate that short hospitalization trajectories exhibit identifiable admission-time patterns, this proxy should be interpreted as an approximation. Accurate next-day discharge forecasting would require temporally structured EHR data and sequential modeling approaches.

FEATURE IMPORTANCE & INTERPRETABILITY

Permutation importance analysis was conducted on the classification model, measuring the reduction in ROC-AUC when each input feature was randomly permuted. This approach captures both direct and indirect feature contributions, including those mediated through learned nonlinear interactions.

Table 3. Top Feature Correlates of Long Stay (Permutation Importance)

Rank	Feature	Importance (AUC Drop)
1	Number of Medications	0.1037
2	Primary Diagnosis (diag_1)	0.0664
3	Number of Lab Procedures	0.0346
4	Number of Procedures	0.0114
5	Secondary Diagnosis (diag_2)	0.0080
6	Tertiary Diagnosis (diag_3)	0.0071
7	Max Glucose Serum	0.0045
8	Age	0.0039
9	Admission Source	0.0035
10	Admission Type	0.0031

Table 3. Permutation importance for the top 10 features from the Neural Network classifier.

BED-OPTIMIZATION SIMULATION

To translate model outputs into concrete operational guidance, a 30-day rolling bed occupancy simulation was developed. The simulation assumed 50 daily patient arrivals, with each patient's LOS drawn from LightGBM predictions on the held-out test set. Bed occupancy was accumulated cumulatively across the simulation window, accounting for staggered arrivals and departures.

The simulation was run under two scenarios: (1) Standard allocation, based on actual predicted LOS values; and (2) What-If scenario, in which 10% of patients were discharged one day earlier than predicted (applicable only to stays greater than one day). This scenario models the effect of targeted early-discharge interventions on peak bed demand.

Results demonstrated that cumulative bed demand stabilized at approximately 200–225 beds by Day 5 under the standard scenario, consistent with the average LOS of 4.40 days and 50 daily arrivals. The what-if scenario produced a modest but sustained reduction in occupancy of approximately 8–12 beds per day — representing approximately 4–5% relief in peak demand. This quantifies the potential operational value of proactive discharge planning for even a small fraction of patients.

The integration of predictive modeling with operational simulation provides a forward-looking planning tool that moves hospitals from reactive crisis management to proactive resource deployment. High-demand periods can be identified several days in advance, enabling targeted staffing, capacity preparation, and discharge coordination.

CONCLUSION

This study demonstrates that machine learning can meaningfully improve both the accuracy and operational utility of hospital LOS prediction. Across regression tasks, LightGBM outperformed all comparators with a test MAE of 1.60 days and R^2 of 0.473 — a level of precision that is clinically actionable for discharge planning and capacity management. For long-stay classification, the Neural Network achieved a ROC-AUC of 0.838, providing a reliable risk signal for proactive bed management.

Several methodological contributions merit emphasis. The patient-level grouped split guards against the optimistic bias introduced by data leakage. Quantile regression with post-hoc calibration extends predictions from point estimates to calibrated intervals with approximately 80% empirical coverage. Subgroup analysis confirmed equitable uncertainty estimation across gender, age, and racial demographics. The bed allocation simulation translates probabilistic forecasts into directly actionable operational scenarios.

Three key research findings emerged: (1) LOS unpredictability is driven by clinical complexity — particularly medication burden, laboratory intensity, and diagnostic count — rather than demographic characteristics; (2) extended hospitalization is most strongly predicted by treatment complexity features at admission, with number of medications as the single most important variable; and (3) early discharge readiness is moderately predictable from encounter-level features (AUC = 0.836), though true next-day forecasting requires temporally structured EHR data.

LightGBM with quantile uncertainty estimation is recommended for operational deployment due to its predictive strength, calibrated uncertainty output, and scalable training characteristics. Future work should explore integration with live EHR systems, real-time dashboards, and multi-hospital validation across diverse patient populations.

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